higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIAL TECHNOLOGY
NQF LEVEL 3

(6021013)

4 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat, clear and in proportion.
5. ALL work you do not want to be marked must be clearly crossed out.
6. Start ALL the questions on a NEW page.
7. Write neatly and legibly.

NCV

QUESTION 1

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–J) next to the question number (1.1–1.8) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1	Glass fibre	A	measure of how well a material can resist changing its shape when something is pressed into it
1.2	Malleability	B	thinking about something carefully in order to understand it
1.3	Thermosetting plastic	C	process also used to give softness and ductility to steel
1.4	Ductility	D	material made of thin fibres of glass
1.5	Normalising	E	allows a material to be drawn out to a long length without breaking
1.6	Shear test	F	when two opposite forces, almost on the same line, cause one part of a structure to slide over another
1.7	Annealing	G	process of heating steel to a particular temperature
1.8	Analyse	H	ability of metal to change shape without fracturing when it is hammered
		I	ability of metal to resist shock or blows
		J	once it is shaped it cannot be changed by heating again

(8 × 1) **[8]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.–2.8) in the ANSWER BOOK.

2.1 An atom is the smallest part of the building blocks of all material around us.

2.2 A molecule is a substance made up of only one type of atom.

2.3 The density of a material is the way its surface reflects light.

2.4 Monomers are materials made up of long chains of identical repeating units.

2.5 Smelting is the process of removing oxygen from metal ore to extract the metal.

2.6 Materials used in a furnace are charge materials.

2.7 Kinetic energy is used for movement.

2.8 A charpy test measures the hardness of a material.

(8 × 1) **[8]**

QUESTION 3

3.1 Define the following physical properties of metals:

3.1.1 Density

3.1.2 Thermal conductivity

3.1.3 Colour

3.1.4 Boiling point

(4 × 2) (8)

3.2 Solid metals have a crystalline structure in which the atoms of each crystal are arranged in a specific geometric pattern called a lattice.

Make a neat 3-dimensional sketch of ONE cell of a body-centred cubic lattice structure and indicate the position of each atom in the structure. (3)

3.3 Give the symbols of the following nonferrous metals as indicated on the periodic table:

3.3.1 Copper

3.3.2 Manganese

3.3.3 Lead

3.3.4 Helium

3.3.5 Aluminum

(5 × 1) (5)

3.4 Metals are alloyed to change their properties to suit certain purposes.

Name FIVE effects that the addition of nickel can have on the properties of a metal. (5)

[21]

QUESTION 4

4.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

4.1.1 Which ONE of the following characteristics best describes a nonmetal?

A Malleable
B Brittle
C Ductile
D Strong

4.1.2 Which ONE of the following non-metals is regarded as the best electrical conductor?

A Bromine
B Graphite
C Fluorine
D Iodine

4.1.3 Which ONE of the following metals is a non-metal?

A Copper
B Carbon
C Iron
D Brass

4.1.4 Nonmetal used in the manufacture of fireworks:

A Arsenic
B Fluorine
C Iodine
D Bromine

4.1.5 Iodine is used in ONE of the following:

A Medicine
B Paint
C Pencils
D Rectifiers

(5 × 1) (5)

4.2 State FOUR properties and FOUR uses of glass reinforced plastic (GRP). (4 × 2) (8)

4.3 Define the following terms in relation to non-metals:

4.3.1 Polymers

4.3.2 Monomers

4.3.3 Elastomers

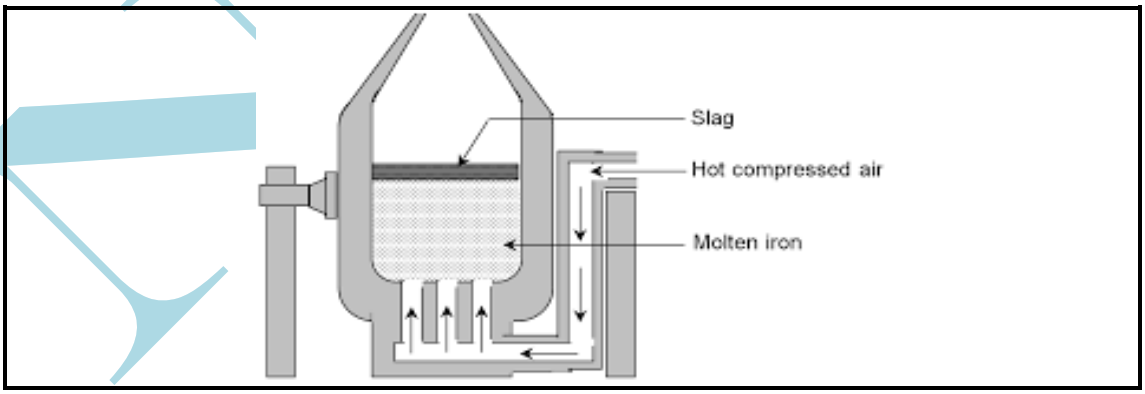
(3 × 2) (6)

4.4 Explain the difference between *synthetic* and *natural* diamonds. (2)
[21]

QUESTION 5

5.1 List FOUR advantages of the cupola furnace. (4)

5.2 Explain the operating principles of the Bessemer converter shown in the diagram below.



BESSEMER CONVERTER

(3 × 2) (6)

5.3 Explain the full annealing process undertaken to soften steel after it has been hardened. (3 × 2) (6)

5.4 Does normalising increase the strength of steel? Answer Yes or No. (1)

5.5 Apart from water, in which liquid is hardened steel quenched immediately after removing it from the heating device? (1)

5.6 Give THREE terms used in connection with the heat treatment of carbon steel. (3)
[21]

QUESTION 6

6.1 Complete the following sentences by using the words in the list below. Write only the missing word next to the question number (6.1.1–6.1.3) in the ANSWER BOOK.

means; 667; rectangular; hardness; 2 000; oxides; circular; edges; quality

6.1.1 The best shape for withstanding torque is a hollow … shaft.

6.1.2 Apart from hardness, another purpose of testing materials is to determine the … of the material.

6.1.3 The test to determine the resistance to penetration of a material is called a … test.

(3 × 1) (3)

6.2 Define *indentation* related to the testing of materials. (2)

6.3 A tensile test was done on a test piece having the following information:

Original length	=	170 mm
Final length	=	185,98 mm
Initial diameter	=	8 mm
Highest force applied	=	25 kN

6.3.1 Calculate the change in length (ΔL) of the test piece in mm. (3)

6.3.2 What is the SI unit for strain? (1)

6.3.3 Calculate the strain. (2)

6.3.4 Calculate the cross-sectional area in m^2, accurate to FIVE decimals. (3)

6.3.5 Calculate the ultimate tensile strength of the material in MPa.

$$\sigma = \frac{F}{A} \qquad \varepsilon = \frac{\Delta L}{L_o} \qquad A = \frac{\pi}{4} d^2$$

(2)

6.4 A test piece was tested using a Brinell hardness tester by using a 10 mm diameter ball. A load was applied, and the diameter of the indentation was measured as 4,54 mm.

6.4.1 Calculate the surface area of the impression in mm^2, accurately to TWO decimals. (2)

6.4.2 If the Brinell hardness number was found to be 204,44, calculate the load applied in Newton.

$$Surface\ area\ of\ impression = \frac{\pi D}{2}\left[D - \sqrt{D^2 - d^2}\right]$$

$$BHN = \frac{Load\ in\ kg}{Surface\ area\ of\ impression}$$

(3)
[21]

TOTAL: 100

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